

Form L1	To be filled and handed over to Head of the Department by the Lecturer-in-charge
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Course Detail Form

Faculty of Science
University of Peradeniya

Course Code	STA 1023/MAT 1063		
Course Title & No. of Credit/s	Introduction to Probability Theory (3 Credits)		
Department	Statistics and Computer Science		
Academic Year & Semester	2024/25 & SII		
Aim/Objective	This course aims to introduce the basic concepts of probability theory.		
Lectures: Day of the week	Monday & Wednesday	Friday	Venue
Time & Venue	11.10 am-12.00 noon	8.00 am -8.50 am	SCS Auditorium
Status compulsory/ Optional	Compulsory		
Pre-requisites	None		
Maximum no. of students allowed	NA		
Primary Instructor's name	Dr. H.T.K. Abeysundara		
Supporting Demonstrator's name	TBA		
Method of evaluation	In-course	20 - 40%	
	End-semester	60 - 80%	

COURSE OUTLINE

Course Learning outcomes:

On successful completion of the course the students should be able to:

- (i) explain basic principles in probability theory.
- (ii) use standard techniques to compute probability measures.
- (iii) evaluate the independence of events.
- (iv) work with discrete and continuous random variables and their distributions.
- (v) explain and compute expectation, variance, moments, and moment generating function of random variables.
translate real world problems into probability models

Week	Lecture Topic
1	CH1-Counting Techniques (Multiplication Rule, Combinations, Permutations)
2	CH2-Probability (Definitions, Calculation, Union, Intersection, Complement, Probability Rules, Venn Diagrams)
3	CH2-Conditional Probability and independency, Law of Total Probability, Bayes' Theorem
4	Problem solving, Examples, Tutorial Discussion,
5	CH3 – Probability Distributions (Definitions, Properties, Cumulative distribution)
6	CH3 – Probability Mass Functions for Discrete random variables
7	CH3 – Probability Density Function for Continuous random variables
8	Problem solving, Examples, Tutorial Discussion
9	CH4 – Mathematical Expectation (Definitions, Calculations)
10	CH4 - Moment Generating Function (Definition, Derivations, Applications), Inequalities
11	CH5 – Special Discrete Distributions (Uniform, Bernoulli, Binomial, Poisson)
12	CH5 – Special Discrete Distributions (Negative Binomial, Geometric, Hypergeometric, Multinomial)
13	Problem solving, Examples, Tutorial Discussion
14	CH6 - Special Continuous Distributions (Uniform, Normal, Exponential Family)
15	Problem solving, Examples, Tutorial Discussion